

RACHANA N P

AI/ML & Edge Systems Engineer

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Third-year Computer Science Engineering student specializing in Edge AI, TinyML, IoT, and embedded systems, with hands-on data science experience. I design end to end — circuit, firmware, on-device model, and the interface someone actually uses — across wearable biosignal devices, environmental sensing systems, and full-stack platforms.

SKILLS

AI / ML: PyTorch, LSTM & time-series, TinyML, reinforcement learning, machine learning fundamentals

Data Science: Python, data analysis, data preprocessing, exploratory data analysis (EDA), data analytics

Edge & Embedded: ESP32 / ESP32-S3, sensor fusion, biosignal acquisition, embedded C++

Full-Stack: Next.js / TypeScript, FastAPI, PostgreSQL, Supabase

Tools: Git, Docker, rapid prototyping, BOM & hardware costing, AI tools

CERTIFICATIONS

Oracle AI Foundations Associate — Oracle

Issued Jan 2026 | Credential ID 324930988OCI25AICFA

Skills: AI and machine learning fundamentals

Nation SkillUp Program — Python, Data Analytics, AI Tools & Git — GeeksforGeeks

Issued Jan 2026

Skills: Python, Git, Data Analytics, AI Tools

EXPERIENCE

AI Engineering Intern — Infosys

June 2026 – Present

- Building an AI Academic Project Mentor platform end to end on Next.js, FastAPI, and Supabase (PostgreSQL, authentication, and row-level security), from system architecture through a shipped onboarding UI.
- Completed Milestone 1 with the team: system and ER diagrams, authentication and project submission flows, and a full presentation deck.

Data Science Intern — Pursuit Future Technologies

Jan 2026 - April 2026

- Strengthened skills in Python, data analysis, machine learning, data preprocessing, and exploratory data analysis (EDA) across applied projects.
- Built a Customer Churn Prediction model and a Blind Auction application.
- Received the Performance of Excellence award in recognition of work during the internship.

PROJECTS

Follicure — In progress

Closed-loop, predictive scalp cooling system for chemotherapy-induced hair loss. Multi-site PPG perfusion sensing with an on-device TinyML model driving zone-addressable cooling, plus a fit-adaptive shell for textured hair. ESP32-S3, MAX30102-class PPG, zone-addressable Peltier control.

VitalDrift — In progress

Wearable cardiac early-warning armband that learns a wearer's personal 14-day biosignal baseline and flags dangerous drift on-device. ESP32-S3, ECG + PPG fusion, on-device quantized LSTM.

ChromaTrack + DyeTag — In progress

Pipe-level spectral fingerprinting system attributing industrial dye discharge to a specific factory using unique fluorescent tracer ratios. AS7265x 18-channel spectral sensor, TinyML.

LakeGuard AI + AquaDrone Sentinel — Shipped, 4th of 147 teams, KSRSAC State-Level Exhibition (Rs. 15,000 prize)

Lake-health monitoring dashboard paired with a floating ESP32 sensor drone; combines live rainfall data with a composite risk model across Bengaluru lakes. React, FastAPI, ESP32.

ACHIEVEMENTS

- 4th place of 147 teams, KSRSAC State-Level Student Innovation Exhibition — Rs. 15,000 cash prize (LakeGuard AI + AquaDrone Sentinel).
- Performance of Excellence award, Data Science Internship at Pursuit Future Technologies.
- Scaler x Meta PyTorch OpenEnv Hackathon — built a Legal Negotiation RL Environment solo, full Docker/OpenEnv CLI pipeline.

EDUCATION

B.E. Computer Science Engineering — Bengaluru

2024 – 2028 | *Specialization focus: Edge AI, TinyML, IoT, and embedded systems*